

John Kyle Cooper

📍 Tokyo, Japan 🌐 Website 🔗 LinkedIn 🐙 GitHub

■ Professional Summary

Research builder and data analyst working at the intersection of human behavior, experimental systems, and AI-assisted scientific workflows. I use programming, data analysis, and pragmatic AI tools to turn complex research questions into experiments, prototypes, analyses, and usable evidence.

■ Research and Technical Strengths

Research Practice: Problem framing, experimental design, human-subject research, scientific writing

Programming and Analysis: Python, MATLAB, R, and JavaScript for research data analysis, automation, scripting, and experimental workflows

Experimental Systems: Audiovisual tasks, stimulus workflows, EEG systems, Unity and VR environments

Agentic AI Workflows: Codex, skills, context and memory management, PRD/ARD/Markdown planning docs, structured grilling, and coding support

Collaboration: Cross-disciplinary teams, participant studies, technical documentation, presentations

■ Education

PhD **KU Leuven**, Biomedical Sciences - Leuven, Belgium Oct 2021 – Oct 2025

MSc **University of Rochester**, Biomedical Engineering - Rochester, New York Aug 2019 – May 2021

BSc **Texas A&M University**, Biomedical Engineering - College Station, Texas May 2019

- Germany Biosciences Program, Bonn, Germany
- Greece Engineering Ethics, Thessaloniki, Greece

■ Research and Technical Experience

IRCN, The University of Tokyo, Project Researcher - Tokyo, Japan Nov 2025 – present

- Study insight, creativity, and problem solving using controlled puzzle tasks and human-centered experimental methods.
- Develop research workflows for turning ambiguous questions about creativity into measurable tasks, participant procedures, and analyzable data.
- Explore AI-supported methods for evaluating creative ideas and supporting research design.

KU Leuven, FWO Strategic Basic Research PhD Fellow - Leuven, Belgium Nov 2021 – Oct 2025

- Built and maintained experiment and data workflows for neuroimaging studies of audiovisual speech perception and hearing outcomes.
- Applied Python, MATLAB, R, and scikit-learn to process time-series data and support forecasting models for speech understanding.
- Redesigned an interactive Unity-based audiovisual experiment to support controlled studies of realistic listening and human-computer interaction conditions.

University of Rochester, Research Assistant - Rochester, New York Aug 2019 – May 2021

- Designed and implemented a Python-based experimental setup for audiovisual neuroimaging research, achieving sub-millisecond video playback timing.
- Supported studies of speech perception and audiovisual congruence by coordinating experimental procedures, participant scheduling, and lab operations.

Texas A&M University, Research Assistant - College Station, Texas Aug 2016 – May 2019

- Built VR and hardware prototypes for optical coherence tomography research, including an Unreal Engine viewer for 3D cochlear scans and fabricated power supply boxes for experimental scanning.

Lynntech Inc. and Enmodes GmbH, Biomedical Engineering Design Teammate - College Station, Texas and Aachen, Germany Jan 2017 – May 2019

- Contributed to early-stage biomedical device design projects, including an orthotic rehabilitation device and a prototype respiratory support system.

■ Selected Publications

- Objectively measuring audiovisual effects in noise using virtual human speakers** Apr 2025
J. K. Cooper, J. Vanthornhout, A. van Wieringen, T. Francart
<https://doi.org/10.1177/23312165251333528> (Trends in Hearing)
- Neural Tracking of Audiovisual Effects in Noise Using Deep Neural Network-Generated Virtual Humans** June 2025
J. K. Cooper, J. Vanthornhout, A. van Wieringen, T. Francart
<https://doi.org/10.1101/2025.06.02.656280> (bioRxiv)
- Musical training does not enhance neural sound encoding at early stages of the auditory system: A large-scale multisite investigation** Sept 2024
K. L. Whiteford, ..., J. K. Cooper, ..., A. J. Oxenham
<https://doi.org/10.1101/2024.09.02.610856> (Nature Communications)

■ Communication and Outreach

- IRCN, The University of Tokyo**, Student Outreach Instructor - Tokyo, Japan 2026
• Led an outreach session on creativity, LLM-supported idea evaluation, and EEG as a method for studying brain activity.
- KU Leuven**, R Programming Teaching Assistant - Leuven, Belgium 2024
• Taught introductory R programming and statistics concepts to audiology students.
- University of Rochester**, Biomaterials Teaching Assistant - Rochester, New York 2020
• Led lab sessions, review sessions, and office hours for biomaterials students.

■ Honors

- FWO Strategic Basic Research Fellowship (2021-2025)
- Belgian American Educational Foundation Fellow (2021)
- Best "New Neuromorph" Award, Telluride Neuromorphic Cognition Engineering Workshop (2022)
- Fulbright Semifinalist for Open Study/Research Award to Belgium (2021)
- BME Graduate Student Teaching Assistant Award Honorable Mention (2020)
- Eagle Scout (2014)